ACADEMIC PROJECT PRESENTATION

BioTwin AI

Digital Twin for Personalized Health Monitoring

PROJECT TEAM

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BACKGROUND & CHALLENGES

Problem Statement

Fragmented Health Data

Sleep, heart rate, and activity metrics remain isolated in separate consumer applications without holistic correlation.

Delayed Interventions

Traditional health monitoring relies on sporadic doctor visits rather than continuous lifestyle monitoring.

Sleep & Glucose Impact

Users lack real-time visibility into how sleep duration directly influences glucose variability and metabolic recovery.

Our Solution: BioTwin AI

Digital Twin Framework

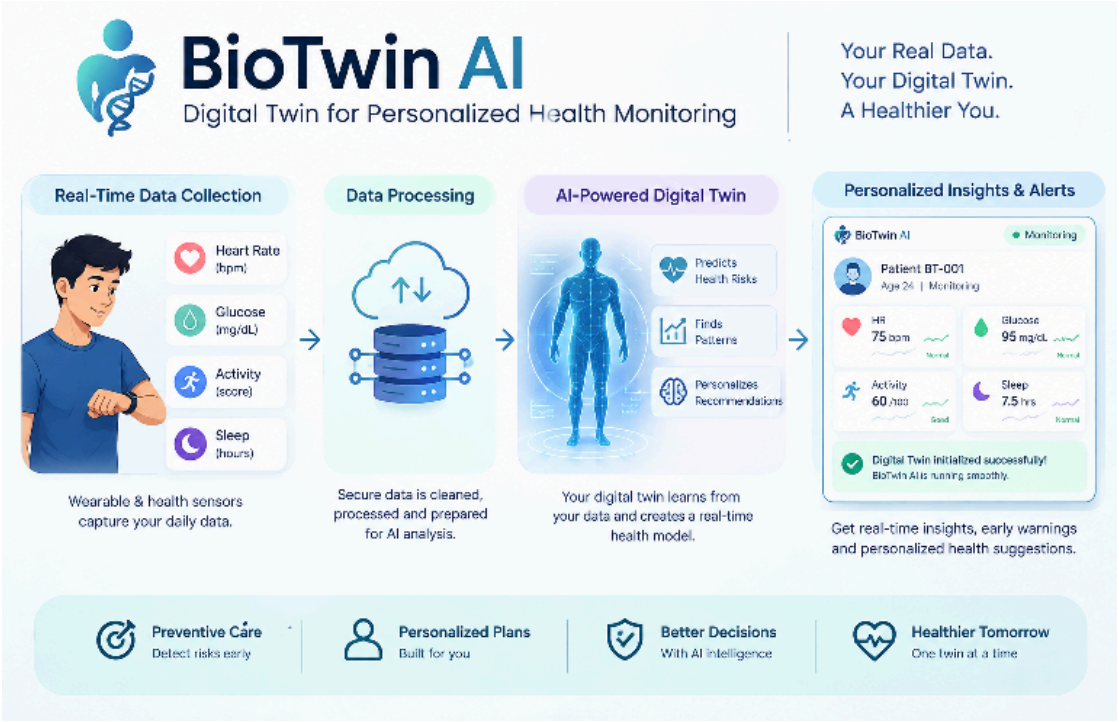
Creates a computational replica of individual patient health states by synchronizing multiple vital streams.

Personalized Monitoring

Tracks sleep patterns, physical activity, heart rate, and blood glucose in a unified analytics layer.

Interactive Central Dashboard

Provides transparent, instant visibility into daily physiological trends for actionable lifestyle adjustments.



DATA PIPELINE & WORKFLOW

System Architecture

STAGE 01

Simulated Patient Data

Generates real-time synthetic physiological health inputs.

STAGE 02

Python Processing

Pandas framework transforms and aggregates patient streams.

STAGE 03

Digital Twin State

Maintains active patient health metrics state vector.

STAGE 04

Streamlit Dashboard

Displays live metrics UI:

- **Sleep Duration**
- **Physical Activity**
- **Heart Rate**
- **Blood Glucose**

TOOLS & LIBRARIES

Technology Stack

Python 3.x

Core Language

Provides backend computational logic, data generation scripts, and system state modeling algorithms.

Streamlit

Frontend Framework

Powers the web user interface, dynamic charts, slider controls, and real-time metric visualizations.

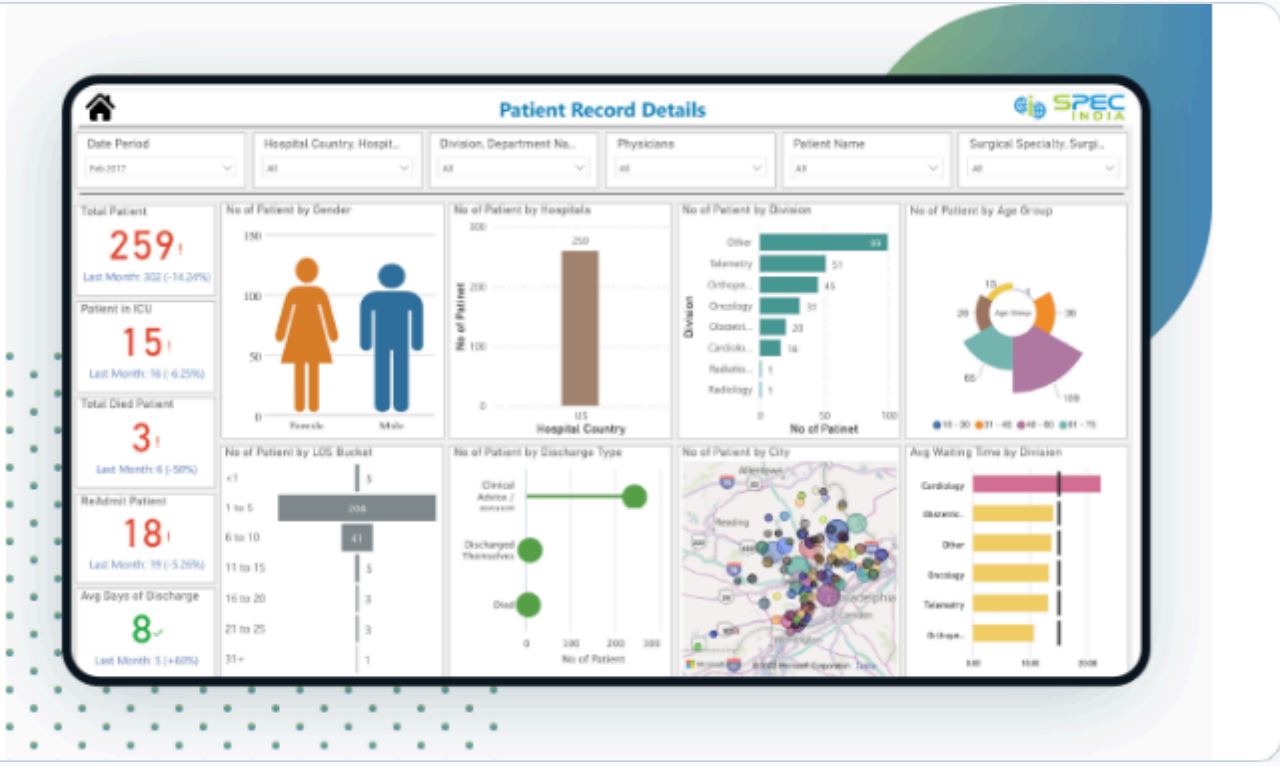
Pandas

Data Engine

Executes tabular data manipulations, time-series data structuring, and metric aggregations.

LIVE DEMONSTRATION

Working Prototype Dashboard



Interactive Controls

Streamlit sidebar allows dynamic patient selection and metric filtering.

Multi-Metric Correlation

Synchronized plots display heart rate spikes alongside glucose fluctuations.

Real-time State Update

Digital Twin status dynamically reflects changes in underlying data tables.

CURRENT STATUS & HONESTY

Results & Scope Limitations

Achieved Capabilities

Simulated Pipeline: Functional data ingestion from rule-based synthetic health datasets.

Interactive UI: Fully responsive Streamlit dashboard demonstrating metric relationships.

Data Orchestration: Pandas framework effectively normalizes multi-parameter patient streams.

Current Prototype Limits

Synthetic Input: Relies exclusively on simulated data; non-integrated with hardware sensors.

No Trained AI Model: Machine learning features are conceptual; predictions are currently rule-based.

Static State: Lacks persistent database memory across web sessions.

NEXT DEVELOPMENT ROADMAP

Future Scope & AI Expansion

Wearable IoT Integration

Direct continuous synchronization with smartwatch APIs (Apple Health, Fitbit, Garmin).



Predictive ML Models

Train LSTM / Transformer neural networks for predictive glucose and fatigue forecasting.



Sleep Pattern Analysis

Deep sleep stage classification and circadian rhythm optimization algorithms.

Thank You! Questions & Feedback Welcome.